PG No. 31 T_{d} $\bar{4}3m$ [cubic] (internal axial quadrupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_{0}^{(2,2)}[g](A_2)$

** symmetry

1

** expression

$$-\frac{\sqrt{5}G_u\left(x^2+y^2-2z^2\right)}{10}+\frac{\sqrt{15}G_v\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{15}G_{xy}xy}{5}+\frac{\sqrt{15}G_{xz}xz}{5}+\frac{\sqrt{15}G_{yz}yz}{5}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_{1,1}^{(2,0)}[g](T_1)$, $\vec{\mathbb{G}}_{1,2}^{(2,0)}[g](T_1)$, $\vec{\mathbb{G}}_{1,3}^{(2,0)}[g](T_1)$

** symmetry

x

y

z

** expression

$$-\frac{\sqrt{10}G_u x}{10}+\frac{\sqrt{30}G_v x}{10}+\frac{\sqrt{30}G_{xy}y}{10}+\frac{\sqrt{30}G_{xz}z}{10}$$

$$-\frac{\sqrt{10}G_u y}{10}-\frac{\sqrt{30}G_v y}{10}+\frac{\sqrt{30}G_{xy}x}{10}+\frac{\sqrt{30}G_{yz}z}{10}$$

$$\frac{\sqrt{10}G_u z}{5}+\frac{\sqrt{30}G_{xz}x}{10}+\frac{\sqrt{30}G_{yz}y}{10}$$

$\vec{\mathbb{G}}_{1,1}^{(2,2)}[g](T_1)$, $\vec{\mathbb{G}}_{1,2}^{(2,2)}[g](T_1)$, $\vec{\mathbb{G}}_{1,3}^{(2,2)}[g](T_1)$

** symmetry

x

y

z

** expression

$$-\frac{3\sqrt{35}G_u x\left(x^2+y^2-4z^2\right)}{70}+\frac{\sqrt{105}G_v x\left(3x^2-7y^2-2z^2\right)}{70}+\frac{\sqrt{105}G_{xy}y\left(4x^2-y^2-z^2\right)}{35}+\frac{\sqrt{105}G_{xz}z\left(4x^2-y^2-z^2\right)}{35}+\frac{\sqrt{105}G_{yz}xyz}{7}$$

$$-\frac{3\sqrt{35}G_u y\left(x^2+y^2-4z^2\right)}{70}+\frac{\sqrt{105}G_v y\left(7x^2-3y^2+2z^2\right)}{70}-\frac{\sqrt{105}G_{xy}x\left(x^2-4y^2+z^2\right)}{35}+\frac{\sqrt{105}G_{xz}xyz}{7}-\frac{\sqrt{105}G_{yz}z\left(x^2-4y^2+z^2\right)}{35}$$

$$-\frac{3\sqrt{35}G_u z\left(3x^2+3y^2-2z^2\right)}{70}+\frac{\sqrt{105}G_v z\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}G_{xy}xyz}{7}-\frac{\sqrt{105}G_{xz}x\left(x^2+y^2-4z^2\right)}{35}-\frac{\sqrt{105}G_{yz}y\left(x^2+y^2-4z^2\right)}{35}$$

$\vec{\mathbb{Q}}_{1,1}^{(2,1)}[g](T_2)$, $\vec{\mathbb{Q}}_{1,2}^{(2,1)}[g](T_2)$, $\vec{\mathbb{Q}}_{1,3}^{(2,1)}[g](T_2)$

** symmetry

x

y

z

** expression

$$-\frac{3\sqrt{10}G_u yz}{10}-\frac{\sqrt{30}G_v yz}{10}+\frac{\sqrt{30}G_{xy}xz}{10}-\frac{\sqrt{30}G_{xz}xy}{10}-\frac{\sqrt{30}G_{yz}\left(y-z\right)\left(y+z\right)}{10}$$

$$\frac{3\sqrt{10}G_u xz}{10}-\frac{\sqrt{30}G_v xz}{10}-\frac{\sqrt{30}G_{xy}yz}{10}+\frac{\sqrt{30}G_{xz}\left(x-z\right)\left(x+z\right)}{10}+\frac{\sqrt{30}G_{yz}xy}{10}$$

$$\frac{\sqrt{30}G_v xy}{5}-\frac{\sqrt{30}G_{xy}\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{30}G_{xz}yz}{10}-\frac{\sqrt{30}G_{yz}xz}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_{2,1}^{(2,-1)}[g](E)$, $\vec{\mathbb{Q}}_{2,2}^{(2,-1)}[g](E)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}G_{xz}y}{2}-\frac{\sqrt{2}G_{yz}x}{2}$$

$$\frac{\sqrt{6}G_{xy}z}{3}-\frac{\sqrt{6}G_{xz}y}{6}-\frac{\sqrt{6}G_{yz}x}{6}$$

$\vec{\mathbb{Q}}^{(2,1)}_{2,1}[g](E)$, $\vec{\mathbb{Q}}^{(2,1)}_{2,2}[g](E)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{5\sqrt{42}G_vxyz}{14}-\frac{5\sqrt{42}G_{xy}z\left(x-y\right)\left(x+y\right)}{28}-\frac{\sqrt{42}G_{xz}y\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{42}G_{yz}x\left(x^2+y^2-4z^2\right)}{28}$$

$$-\frac{5\sqrt{42}G_uxyz}{14}+\frac{\sqrt{14}G_{xy}z\left(3x^2+3y^2-2z^2\right)}{28}-\frac{\sqrt{14}G_{xz}y\left(9x^2-y^2-6z^2\right)}{28}+\frac{\sqrt{14}G_{yz}x\left(x^2-9y^2+6z^2\right)}{28}$$

$\vec{\mathbb{G}}^{(2,-2)}_{2,1}[g](E)$, $\vec{\mathbb{G}}^{(2,-2)}_{2,2}[g](E)$
** symmetry

$$-\frac{\sqrt{3}x^2}{2}+\frac{\sqrt{3}y^2}{2}$$

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-G_v$$

$$G_u$$

$\vec{\mathbb{G}}^{(2,0)}_{2,1}[g](E)$, $\vec{\mathbb{G}}^{(2,0)}_{2,2}[g](E)$
** symmetry

$$-\frac{\sqrt{3}x^2}{2}+\frac{\sqrt{3}y^2}{2}$$

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{42}G_u\left(x-y\right)\left(x+y\right)}{14}-\frac{\sqrt{14}G_v\left(x^2+y^2-2z^2\right)}{14}-\frac{3\sqrt{14}G_{xz}xz}{14}+\frac{3\sqrt{14}G_{yz}yz}{14}$$

$$-\frac{\sqrt{14}G_u\left(x^2+y^2-2z^2\right)}{14}-\frac{\sqrt{42}G_v\left(x-y\right)\left(x+y\right)}{14}-\frac{\sqrt{42}G_{xy}xy}{7}+\frac{\sqrt{42}G_{xz}xz}{14}+\frac{\sqrt{42}G_{yz}yz}{14}$$

$\vec{\mathbb{G}}^{(2,2)}_{2,1}[g](E)$, $\vec{\mathbb{G}}^{(2,2)}_{2,2}[g](E)$
** symmetry

$$-\frac{\sqrt{3}x^2}{2}+\frac{\sqrt{3}y^2}{2}$$

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{5\sqrt{42}G_u\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{168}-\frac{\sqrt{14}G_v\left(19x^4-102x^2y^2-12x^2z^2+19y^4-12y^2z^2+4z^4\right)}{168}$$
$$-\frac{5\sqrt{14}G_{xy}xy\left(x-y\right)\left(x+y\right)}{12}-\frac{5\sqrt{14}G_{xz}xz\left(5x^2-9y^2-2z^2\right)}{84}-\frac{5\sqrt{14}G_{yz}yz\left(9x^2-5y^2+2z^2\right)}{84}$$

$$\frac{\sqrt{14}G_u\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{56}-\frac{5\sqrt{42}G_v\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{168}$$
$$-\frac{5\sqrt{42}G_{xy}xy\left(x^2+y^2-6z^2\right)}{84}-\frac{5\sqrt{42}G_{xz}xz\left(3x^2+3y^2-4z^2\right)}{84}-\frac{5\sqrt{42}G_{yz}yz\left(3x^2+3y^2-4z^2\right)}{84}$$

$\vec{\mathbb{G}}_{2,1}^{(2,-2)}[g](T_1)$, $\vec{\mathbb{G}}_{2,2}^{(2,-2)}[g](T_1)$, $\vec{\mathbb{G}}_{2,3}^{(2,-2)}[g](T_1)$
** symmetry

$\sqrt{3}yz$

$\sqrt{3}xz$

$\sqrt{3}xy$

** expression

G_{yz}

G_{xz}

G_{xy}

$\vec{\mathbb{G}}_{2,1}^{(2,0)}[g](T_1)$, $\vec{\mathbb{G}}_{2,2}^{(2,0)}[g](T_1)$, $\vec{\mathbb{G}}_{2,3}^{(2,0)}[g](T_1)$
** symmetry

$\sqrt{3}yz$

$\sqrt{3}xz$

$\sqrt{3}xy$

** expression

$$\frac{\sqrt{42}G_uyz}{14}-\frac{3\sqrt{14}G_vyz}{14}+\frac{3\sqrt{14}G_{xy}xz}{14}+\frac{3\sqrt{14}G_{xz}xy}{14}-\frac{\sqrt{14}G_{yz}\left(2x^2-y^2-z^2\right)}{14}$$

$$\frac{\sqrt{42}G_uxz}{14}+\frac{3\sqrt{14}G_vxz}{14}+\frac{3\sqrt{14}G_{xy}yz}{14}+\frac{\sqrt{14}G_{xz}\left(x^2-2y^2+z^2\right)}{14}+\frac{3\sqrt{14}G_{yz}xy}{14}$$

$$-\frac{\sqrt{42}G_uxy}{7}+\frac{\sqrt{14}G_{xy}\left(x^2+y^2-2z^2\right)}{14}+\frac{3\sqrt{14}G_{xz}yz}{14}+\frac{3\sqrt{14}G_{yz}xz}{14}$$

$\vec{\mathbb{G}}_{2,1}^{(2,2)}[g](T_1)$, $\vec{\mathbb{G}}_{2,2}^{(2,2)}[g](T_1)$, $\vec{\mathbb{G}}_{2,3}^{(2,2)}[g](T_1)$
** symmetry

$\sqrt{3}yz$

$\sqrt{3}xz$

$\sqrt{3}xy$

** expression

$$-\frac{5\sqrt{42}G_uyz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{5\sqrt{14}G_vyz\left(9x^2-5y^2+2z^2\right)}{84}-\frac{5\sqrt{14}G_{xy}xz\left(x^2-6y^2+z^2\right)}{42}$$
$$-\frac{5\sqrt{14}G_{xz}xy\left(x^2+y^2-6z^2\right)}{42}+\frac{\sqrt{14}G_{yz}\left(x^4-3x^2y^2-3x^2z^2-4y^4+27y^2z^2-4z^4\right)}{42}$$

$$-\frac{5\sqrt{42}G_uxz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{5\sqrt{14}G_vxz\left(5x^2-9y^2-2z^2\right)}{84}+\frac{5\sqrt{14}G_{xy}yz\left(6x^2-y^2-z^2\right)}{42}$$
$$-\frac{\sqrt{14}G_{xz}\left(4x^4+3x^2y^2-27x^2z^2-y^4+3y^2z^2+4z^4\right)}{42}-\frac{5\sqrt{14}G_{yz}xy\left(x^2+y^2-6z^2\right)}{42}$$

$$-\frac{5\sqrt{42}G_u xy\left(x^2+y^2-6z^2\right)}{84}+\frac{5\sqrt{14}G_v xy\left(x-y\right)\left(x+y\right)}{12}-\frac{\sqrt{14}G_{xy}\left(4x^4-27x^2y^2+3x^2z^2+4y^4+3y^2z^2-z^4\right)}{42}$$
$$+\frac{5\sqrt{14}G_{xz}yz\left(6x^2-y^2-z^2\right)}{42}-\frac{5\sqrt{14}G_{yz}xz\left(x^2-6y^2+z^2\right)}{42}$$

$\vec{\mathbb{Q}}_{2,1}^{(2,-1)}[g](T_2)$, $\vec{\mathbb{Q}}_{2,2}^{(2,-1)}[g](T_2)$, $\vec{\mathbb{Q}}_{2,3}^{(2,-1)}[g](T_2)$

** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}G_u x}{2}+\frac{\sqrt{6}G_v x}{6}+\frac{\sqrt{6}G_{xy}y}{6}-\frac{\sqrt{6}G_{xz}z}{6}$$

$$-\frac{\sqrt{2}G_u y}{2}+\frac{\sqrt{6}G_v y}{6}-\frac{\sqrt{6}G_{xy}x}{6}+\frac{\sqrt{6}G_{yz}z}{6}$$

$$-\frac{\sqrt{6}G_v z}{3}+\frac{\sqrt{6}G_{xz}x}{6}-\frac{\sqrt{6}G_{yz}y}{6}$$

$\vec{\mathbb{Q}}_{2,1}^{(2,1)}[g](T_2)$, $\vec{\mathbb{Q}}_{2,2}^{(2,1)}[g](T_2)$, $\vec{\mathbb{Q}}_{2,3}^{(2,1)}[g](T_2)$

** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{42}G_u x\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{14}G_v x\left(x^2-9y^2+6z^2\right)}{28}-\frac{\sqrt{14}G_{xy}y\left(3x^2-2y^2+3z^2\right)}{14}+\frac{\sqrt{14}G_{xz}z\left(3x^2+3y^2-2z^2\right)}{14}$$

$$\frac{\sqrt{42}G_u y\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{14}G_v y\left(9x^2-y^2-6z^2\right)}{28}-\frac{\sqrt{14}G_{xy}x\left(2x^2-3y^2-3z^2\right)}{14}-\frac{\sqrt{14}G_{yz}z\left(3x^2+3y^2-2z^2\right)}{14}$$

$$\frac{5\sqrt{42}G_u z\left(x-y\right)\left(x+y\right)}{28}-\frac{\sqrt{14}G_v z\left(3x^2+3y^2-2z^2\right)}{28}+\frac{\sqrt{14}G_{xz}x\left(2x^2-3y^2-3z^2\right)}{14}+\frac{\sqrt{14}G_{yz}y\left(3x^2-2y^2+3z^2\right)}{14}$$

* Harmonics for rank 3

$\vec{\mathbb{Q}}_{3}^{(2,-1)}[g](A_1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{6}G_u\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{2}G_v\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{Q}}_{3}^{(2,1)}[g](A_1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{\sqrt{3}G_u\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{6}-\frac{G_v\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{6}$$
$$-\frac{7G_{xy}xy\left(x-y\right)\left(x+y\right)}{6}+\frac{7G_{xz}xz\left(x-z\right)\left(x+z\right)}{6}-\frac{7G_{yz}yz\left(y-z\right)\left(y+z\right)}{6}$$

$\vec{\mathbb{G}}_{3}^{(2,-2)}[g](A_2)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{3}G_{xy}z}{3}+\frac{\sqrt{3}G_{xz}y}{3}+\frac{\sqrt{3}G_{yz}x}{3}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](A_2)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{3}G_{xy}z\left(3x^2+3y^2-2z^2\right)}{6}+\frac{\sqrt{3}G_{xz}y\left(3x^2-2y^2+3z^2\right)}{6}-\frac{\sqrt{3}G_{yz}x\left(2x^2-3y^2-3z^2\right)}{6}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](A_2)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{21\sqrt{22}G_uxyz\left(x^2+y^2-2z^2\right)}{44}+\frac{21\sqrt{66}G_vxyz\left(x-y\right)\left(x+y\right)}{44}-\frac{\sqrt{66}G_{xy}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{66}$$
$$-\frac{\sqrt{66}G_{xz}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{66}+\frac{\sqrt{66}G_{yz}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{66}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,-1)}[g](T_1)$, $\vec{\mathbb{Q}}_{3,2}^{(2,-1)}[g](T_1)$, $\vec{\mathbb{Q}}_{3,3}^{(2,-1)}[g](T_1)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{6}G_uyz}{4}-\frac{3\sqrt{2}G_vyz}{4}+\frac{\sqrt{2}G_{yz}\left(2x^2-y^2-z^2\right)}{4}$$

$$\frac{\sqrt{6}G_uxz}{4}+\frac{3\sqrt{2}G_vxz}{4}-\frac{\sqrt{2}G_{xz}\left(x^2-2y^2+z^2\right)}{4}$$

$$-\frac{\sqrt{6}G_uxy}{2}-\frac{\sqrt{2}G_{xy}\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,1)}[g](T_1)$, $\vec{\mathbb{Q}}_{3,2}^{(2,1)}[g](T_1)$, $\vec{\mathbb{Q}}_{3,3}^{(2,1)}[g](T_1)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_uyz\left(12x^2-9y^2+5z^2\right)}{12}-\frac{G_vyz\left(36x^2+y^2-13z^2\right)}{12}+\frac{7G_{xy}xz\left(2x^2-3y^2-z^2\right)}{12}$$
$$+\frac{7G_{xz}xy\left(2x^2-y^2-3z^2\right)}{12}-\frac{G_{yz}\left(4x^4-12x^2y^2-12x^2z^2+5y^4-18y^2z^2+5z^4\right)}{12}$$

$$-\frac{\sqrt{3}G_uxz\left(9x^2-12y^2-5z^2\right)}{12}+\frac{G_vxz\left(x^2+36y^2-13z^2\right)}{12}-\frac{7G_{xy}yz\left(3x^2-2y^2+z^2\right)}{12}$$
$$-\frac{G_{xz}\left(5x^4-12x^2y^2-18x^2z^2+4y^4-12y^2z^2+5z^4\right)}{12}-\frac{7G_{yz}xy\left(x^2-2y^2+3z^2\right)}{12}$$

$$\frac{\sqrt{3}G_u xy\left(x^2+y^2-6z^2\right)}{3}+\frac{7G_v xy\left(x-y\right)\left(x+y\right)}{6}-\frac{G_{xy}\left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{12}$$
$$-\frac{7G_{xz}yz\left(3x^2+y^2-2z^2\right)}{12}-\frac{7G_{yz}xz\left(x^2+3y^2-2z^2\right)}{12}$$

$\vec{\mathbb{G}}_{3,1}^{(2,-2)}[g](T_1)$, $\vec{\mathbb{G}}_{3,2}^{(2,-2)}[g](T_1)$, $\vec{\mathbb{G}}_{3,3}^{(2,-2)}[g](T_1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_u x}{10}+\frac{3\sqrt{5}G_v x}{10}-\frac{\sqrt{5}G_{xy}y}{5}-\frac{\sqrt{5}G_{xz}z}{5}$$

$$-\frac{\sqrt{15}G_u y}{10}-\frac{3\sqrt{5}G_v y}{10}-\frac{\sqrt{5}G_{xy}x}{5}-\frac{\sqrt{5}G_{yz}z}{5}$$

$$\frac{\sqrt{15}G_u z}{5}-\frac{\sqrt{5}G_{xz}x}{5}-\frac{\sqrt{5}G_{yz}y}{5}$$

$\vec{\mathbb{G}}_{3,1}^{(2,0)}[g](T_1)$, $\vec{\mathbb{G}}_{3,2}^{(2,0)}[g](T_1)$, $\vec{\mathbb{G}}_{3,3}^{(2,0)}[g](T_1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_u x\left(4x^2-21y^2+9z^2\right)}{60}+\frac{\sqrt{5}G_v x\left(4x^2-y^2-11z^2\right)}{20}+\frac{\sqrt{5}G_{xy}y\left(4x^2-y^2-z^2\right)}{20}+\frac{\sqrt{5}G_{xz}z\left(4x^2-y^2-z^2\right)}{20}-\sqrt{5}G_{yz}xyz$$

$$\frac{\sqrt{15}G_u y\left(21x^2-4y^2-9z^2\right)}{60}+\frac{\sqrt{5}G_v y\left(x^2-4y^2+11z^2\right)}{20}-\frac{\sqrt{5}G_{xy}x\left(x^2-4y^2+z^2\right)}{20}-\sqrt{5}G_{xz}xyz-\frac{\sqrt{5}G_{yz}z\left(x^2-4y^2+z^2\right)}{20}$$

$$-\frac{\sqrt{15}G_u z\left(3x^2+3y^2-2z^2\right)}{15}-\frac{\sqrt{5}G_v z\left(x-y\right)\left(x+y\right)}{2}-\sqrt{5}G_{xy}xyz-\frac{\sqrt{5}G_{xz}x\left(x^2+y^2-4z^2\right)}{20}-\frac{\sqrt{5}G_{yz}y\left(x^2+y^2-4z^2\right)}{20}$$

$\vec{\mathbb{G}}_{3,1}^{(2,2)}[g](T_1)$, $\vec{\mathbb{G}}_{3,2}^{(2,2)}[g](T_1)$, $\vec{\mathbb{G}}_{3,3}^{(2,2)}[g](T_1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{330}G_u x\left(4x^4+x^2y^2-41x^2z^2-3y^4+15y^2z^2+18z^4\right)}{264}+\frac{\sqrt{110}G_v x\left(4x^4-27x^2y^2-13x^2z^2+11y^4+15y^2z^2+4z^4\right)}{88}$$
$$+\frac{\sqrt{110}G_{xy}y\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{44}+\frac{\sqrt{110}G_{xz}z\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{44}+\frac{7\sqrt{110}G_{yz}xyz\left(2x^2-y^2-z^2\right)}{44}$$

$$\frac{\sqrt{330}G_u y\left(3x^4-x^2y^2-15x^2z^2-4y^4+41y^2z^2-18z^4\right)}{264}-\frac{\sqrt{110}G_v y\left(11x^4-27x^2y^2+15x^2z^2+4y^4-13y^2z^2+4z^4\right)}{88}$$
$$+\frac{\sqrt{110}G_{xy}x\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{44}-\frac{7\sqrt{110}G_{xz}xyz\left(x^2-2y^2+z^2\right)}{44}+\frac{\sqrt{110}G_{yz}z\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{44}$$

$$\frac{\sqrt{330}G_u z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264}-\frac{7\sqrt{110}G_v z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}-\frac{7\sqrt{110}G_{xy}xyz\left(x^2+y^2-2z^2\right)}{44}$$
$$+\frac{\sqrt{110}G_{xz}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}+\frac{\sqrt{110}G_{yz}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,-1)}[g](T_2)$, $\vec{\mathbb{Q}}_{3,2}^{(2,-1)}[g](T_2)$, $\vec{\mathbb{Q}}_{3,3}^{(2,-1)}[g](T_2)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{3\sqrt{10}G_u yz}{20}+\frac{\sqrt{30}G_v yz}{20}+\frac{\sqrt{30}G_{xy}xz}{5}-\frac{\sqrt{30}G_{xz}xy}{5}+\frac{\sqrt{30}G_{yz}\left(y-z\right)\left(y+z\right)}{20}$$

$$-\frac{3\sqrt{10}G_u xz}{20}+\frac{\sqrt{30}G_v xz}{20}-\frac{\sqrt{30}G_{xy}yz}{5}-\frac{\sqrt{30}G_{xz}\left(x-z\right)\left(x+z\right)}{20}+\frac{\sqrt{30}G_{yz}xy}{5}$$

$$-\frac{\sqrt{30}G_v xy}{10}+\frac{\sqrt{30}G_{xy}\left(x-y\right)\left(x+y\right)}{20}+\frac{\sqrt{30}G_{xz}yz}{5}-\frac{\sqrt{30}G_{yz}xz}{5}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,1)}[g](T_2)$, $\vec{\mathbb{Q}}_{3,2}^{(2,1)}[g](T_2)$, $\vec{\mathbb{Q}}_{3,3}^{(2,1)}[g](T_2)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{5}G_u yz\left(6x^2-y^2-z^2\right)}{4}-\frac{\sqrt{15}G_v yz\left(6x^2-y^2-z^2\right)}{12}+\frac{\sqrt{15}G_{xy}xz\left(4x^2-3y^2-3z^2\right)}{12}$$
$$-\frac{\sqrt{15}G_{xz}xy\left(4x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{15}G_{yz}\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{12}$$

$$-\frac{\sqrt{5}G_u xz\left(x^2-6y^2+z^2\right)}{4}+\frac{\sqrt{15}G_v xz\left(x^2-6y^2+z^2\right)}{12}+\frac{\sqrt{15}G_{xy}yz\left(3x^2-4y^2+3z^2\right)}{12}$$
$$-\frac{\sqrt{15}G_{xz}\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{12}-\frac{\sqrt{15}G_{yz}xy\left(3x^2-4y^2+3z^2\right)}{12}$$

$$-\frac{\sqrt{15}G_v xy\left(x^2+y^2-6z^2\right)}{6}+\frac{\sqrt{15}G_{xy}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}-\frac{\sqrt{15}G_{xz}yz\left(3x^2+3y^2-4z^2\right)}{12}+\frac{\sqrt{15}G_{yz}xz\left(3x^2+3y^2-4z^2\right)}{12}$$

$\vec{\mathbb{G}}_{3,1}^{(2,-2)}[g](T_2)$, $\vec{\mathbb{G}}_{3,2}^{(2,-2)}[g](T_2)$, $\vec{\mathbb{G}}_{3,3}^{(2,-2)}[g](T_2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{G_u x}{2}-\frac{\sqrt{3}G_v x}{6}+\frac{\sqrt{3}G_{xy}y}{3}-\frac{\sqrt{3}G_{xz}z}{3}$$

$$\frac{G_u y}{2}-\frac{\sqrt{3}G_v y}{6}-\frac{\sqrt{3}G_{xy}x}{3}+\frac{\sqrt{3}G_{yz}z}{3}$$

$$\frac{\sqrt{3}G_v z}{3} + \frac{\sqrt{3}G_{xz} x}{3} - \frac{\sqrt{3}G_{yz} y}{3}$$

$\vec{\mathbb{G}}^{(2,0)}_{3,1}[g](T_2)$, $\vec{\mathbb{G}}^{(2,0)}_{3,2}[g](T_2)$, $\vec{\mathbb{G}}^{(2,0)}_{3,3}[g](T_2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2 y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2 z}{2} - \frac{\sqrt{15}y^2 z}{2}$$

** expression

$$\frac{G_u x\left(2x^2 - 3y^2 - 3z^2\right)}{4} + \frac{\sqrt{3}G_v x\left(2x^2 - 3y^2 - 3z^2\right)}{12} + \frac{\sqrt{3}G_{xy} y\left(6x^2 + y^2 - 9z^2\right)}{12} - \frac{\sqrt{3}G_{xz} z\left(6x^2 - 9y^2 + z^2\right)}{12}$$

$$\frac{G_u y\left(3x^2 - 2y^2 + 3z^2\right)}{4} - \frac{\sqrt{3}G_v y\left(3x^2 - 2y^2 + 3z^2\right)}{12} - \frac{\sqrt{3}G_{xy} x\left(x^2 + 6y^2 - 9z^2\right)}{12} - \frac{\sqrt{3}G_{yz} z\left(9x^2 - 6y^2 - z^2\right)}{12}$$

$$\frac{\sqrt{3}G_v z\left(3x^2 + 3y^2 - 2z^2\right)}{6} + \frac{\sqrt{3}G_{xz} x\left(x^2 - 9y^2 + 6z^2\right)}{12} + \frac{\sqrt{3}G_{yz} y\left(9x^2 - y^2 - 6z^2\right)}{12}$$

$\vec{\mathbb{G}}^{(2,2)}_{3,1}[g](T_2)$, $\vec{\mathbb{G}}^{(2,2)}_{3,2}[g](T_2)$, $\vec{\mathbb{G}}^{(2,2)}_{3,3}[g](T_2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2 y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2 z}{2} - \frac{\sqrt{15}y^2 z}{2}$$

** expression

$$-\frac{\sqrt{22}G_u x\left(2x^4 + 11x^2y^2 - 31x^2z^2 + 9y^4 - 87y^2z^2 + 30z^4\right)}{88} - \frac{\sqrt{66}G_v x\left(2x^4 - 73x^2y^2 + 53x^2z^2 + 51y^4 - 87y^2z^2 - 12z^4\right)}{264}$$
$$- \frac{\sqrt{66}G_{xy} y\left(12x^4 - 46x^2y^2 + 66x^2z^2 + 5y^4 - 4y^2z^2 - 9z^4\right)}{132}$$
$$+ \frac{\sqrt{66}G_{xz} z\left(12x^4 + 66x^2y^2 - 46x^2z^2 - 9y^4 - 4y^2z^2 + 5z^4\right)}{132} + \frac{21\sqrt{66}G_{yz} xyz\left(y - z\right)\left(y + z\right)}{44}$$

$$\frac{\sqrt{22}G_u y\left(9x^4 + 11x^2y^2 - 87x^2z^2 + 2y^4 - 31y^2z^2 + 30z^4\right)}{88} - \frac{\sqrt{66}G_v y\left(51x^4 - 73x^2y^2 - 87x^2z^2 + 2y^4 + 53y^2z^2 - 12z^4\right)}{264}$$
$$+ \frac{\sqrt{66}G_{xy} x\left(5x^4 - 46x^2y^2 - 4x^2z^2 + 12y^4 + 66y^2z^2 - 9z^4\right)}{132} - \frac{21\sqrt{66}G_{xz} xyz\left(x - z\right)\left(x + z\right)}{44}$$
$$+ \frac{\sqrt{66}G_{yz} z\left(9x^4 - 66x^2y^2 + 4x^2z^2 - 12y^4 + 46y^2z^2 - 5z^4\right)}{132}$$

$$-\frac{21\sqrt{22}G_u z\left(x - y\right)\left(x + y\right)\left(x^2 + y^2 - 2z^2\right)}{88} + \frac{\sqrt{66}G_v z\left(39x^4 - 174x^2y^2 - 20x^2z^2 + 39y^4 - 20y^2z^2 + 4z^4\right)}{264} + \frac{21\sqrt{66}G_{xy} xyz\left(x - y\right)\left(x + y\right)}{44}$$
$$- \frac{\sqrt{66}G_{xz} x\left(5x^4 - 4x^2y^2 - 46x^2z^2 - 9y^4 + 66y^2z^2 + 12z^4\right)}{132} - \frac{\sqrt{66}G_{yz} y\left(9x^4 + 4x^2y^2 - 66x^2z^2 - 5y^4 + 46y^2z^2 - 12z^4\right)}{132}$$